Data Analyst Portfolio

Uber Trip Analysis

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Data Overview

Field name	Type	Description
start_date	DATE	Tanggal perjalanan dimulai
start_hour	FLOAT	Jam dimulainya perjalanan
start_minute	FLOAT	Menit saat perjalanan dimulai
end_date	DATE	Tanggal perjalanan selesai
end_hour	FLOAT	Jam perjalanan selesai
end_minute	FLOAT	Menit saat perjalanan selesai
month	STRING	Bulan perjalanan (format teks, misal: "January")
year	INTEGER	Tahun perjalanan
CATEGORY	STRING	Kategori pengguna (misalnya: "Business", "Personal")
START	STRING	Titik awal lokasi perjalanan
STOP	STRING	Titik akhir lokasi perjalanan
MILES	FLOAT	Jarak tempuh perjalanan dalam mil
PURPOSE	STRING	Tujuan perjalanan (misal: Meeting, Errand, etc.)
duration	FLOAT	Lama perjalanan dalam menit
speed	FLOAT	Kecepatan rata-rata selama perjalanan (mil per jam)
day	STRING	Hari dalam seminggu saat perjalanan terjadi (misal: "Monday")
day_time	STRING	Waktu umum dalam hari (misal: "Morning", "Afternoon", dst.)

Number of Columns: 17
Number of Rows: 1141

Uber trip data was used for this analysis. The dataset was already clean and well-prepared, allowing the analysis to proceed without additional data cleaning.

Tools Used:

- Google BigQuery
- Google Sheets

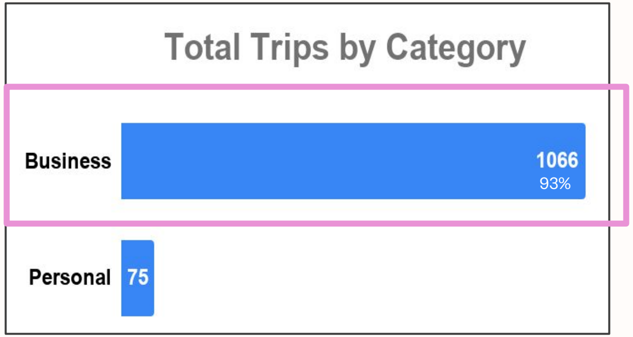
Analysis Objectives

Uber is a technology company that provides on-demand transportation services through its mobile application. As a leading player in the ride-hailing industry, understanding trip demand patterns and travel behavior can help support operational planning and service optimization.

1. **Analyze overall trip patterns** using key metrics such as total trips, average trip duration, and average travelled distance, segmented by user category and day type (weekdays vs. weekend).
2. **Identify peak hours** to better understand customer demand patterns and service utilization.
3. **Evaluate trip efficiency** based on average speed (km/h) to identify periods with lower travel efficiency.

Note: Distance metrics are converted from miles to kilometers (km), and speed metrics are converted to kilometers per hour (km/h).

Trip Patterns by Category



Total Trip:
1.141 trips

	Weekdays			
category	Total Trip	Avg Duration (Minutes)	Avg Duration (Km)	Avg Duration (Km/h)
Business	796	22.64	25.2	71.29
Personal	52	15.46	15.49	52.35

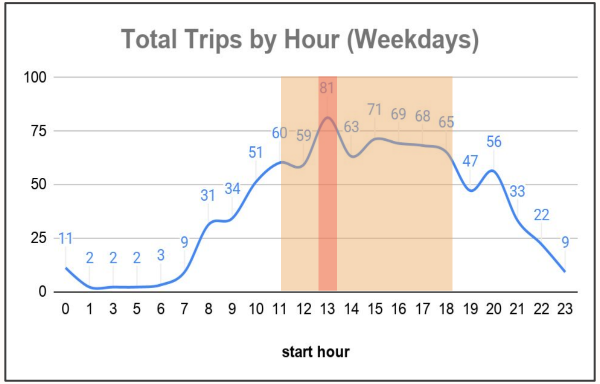
	Weekend			
category	Total Trip	Avg Duration (Minutes)	Avg Duration (Km)	Avg Duration (Km/h)
Business	270	23.76	29.75	66.94
Personal	23	31.78	44.53	58.3

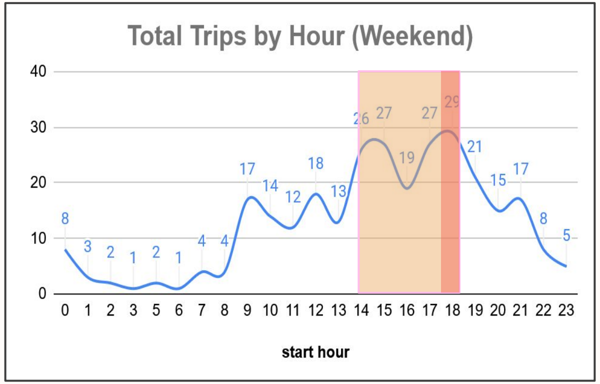
1. **Uber trips were dominated by the Business category** (93%), indicating that the majority of trips were made for business-related purposes.
2. **Average trip distance and duration were higher on weekends**, particularly for Personal trips, with an average distance of 44.5 km and an average duration of 31.7 minutes.

Recommendations:

- Prioritize Business user retention, as this segment accounts for 93% of total trips.
- Explore opportunities within Personal weekend trips, as this segment records the highest average trip distance and duration.

Peak Hour Analysis





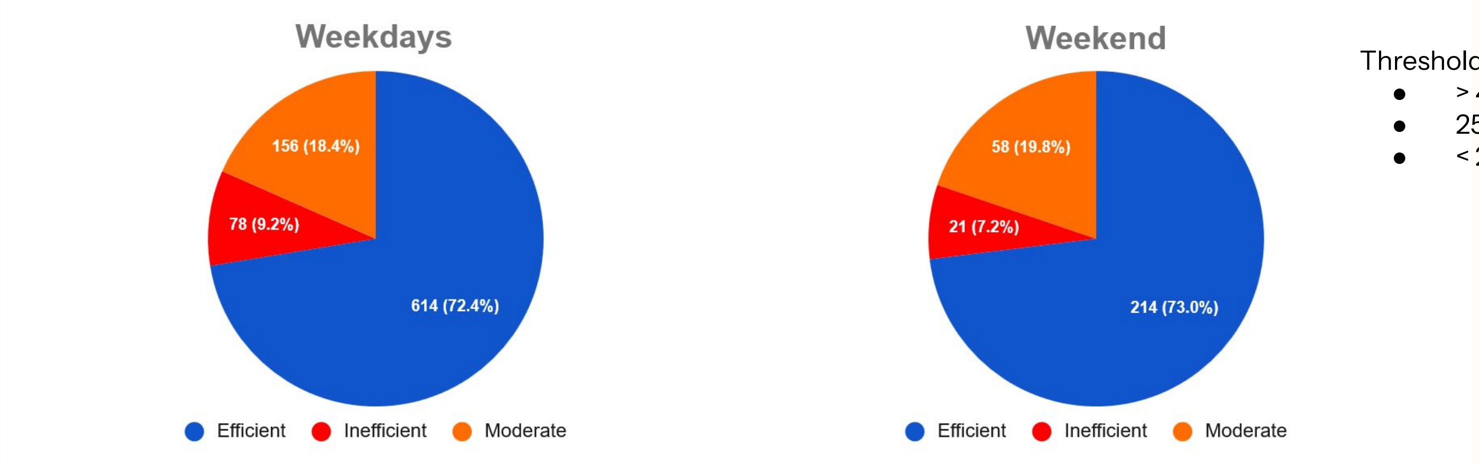
Day	Start hour	Count trip
Friday	11:00	24
Friday	13:00	24
Tuesday	13:00	18
Tuesday	17:00	18
Wednesday	17:00	18
Monday	14:00	17
Friday	15:00	17
Saturday	17:00	17
Thursday	18:00	17
Sunday	18:00	17

1. **On weekdays, trip demand peaks at 13:00 (81 trips)**. Trip activity remains relatively high between 11:00 and 18:00, indicating sustained demand throughout working hours and the afternoon period.
2. **On weekends,** trip demand is highest between 14:00 and 18:00, **reaching its peak at 18:00 (29 trips).**
3. Friday at 11:00 and 13:00 records the highest trip volume (24 trips).
4. Demand is concentrated during midday to afternoon hours (11:00–18:00).

Recommendations:

- Optimize driver availability during peak demand periods (11:00–18:00) to better align supply with customer demand.
- Consider targeted driver incentive programs during peak hours to help maintain adequate driver coverage and service quality.

Trip Efficiency Analysis



Threshold kecepatan:

- > 40 km/h = Efficient
- 25-40 km/h = Moderate
- < 25 km/h = Inefficient

1. **Both weekdays and weekends are predominantly efficient**, with approximately 73% of trips classified as efficient. However, weekdays show a slightly higher proportion of inefficient trips (9.2% vs. 7.2%).
2. Weekends have a higher share of moderate-efficiency trips (19.8%) compared to weekdays (18.4%).
3. Overall, trip efficiency patterns remain relatively stable across both day types.

Trip Efficiency Analysis

Weekdays

Start Hour	Inefficient Trips Count	Total Trips	Inefficient Percentage
16	12	69	17.39
13	12	81	14.81
19	6	47	12.77
14	8	63	12.7
15	7	71	9.86

Purpose	Inefficient Trips Count	Total Trips	Inefficient Percentage
Temporary Site	20	155	12.9
Meal/Entertain	22	180	12.22

Weekend

Start Hour	Inefficient Trips Count	Total Trips	Inefficient Percentage
13	3	13	23.08
15	4	27	14.81
10	2	14	14.29
17	3	27	11.11
9	1	17	5.88

Purpose	Inefficient Trips Count	Total Trips	Inefficient Percentage
Errand/Supplies	8	50	16
Meeting	9	85	10.59

1. **On weekdays, the highest proportions of inefficient trips occur at 13:00 (14.8%) and 16:00 (17.4%). Temporary Site and Meal/Entertain** are the most common trip purposes during these periods, suggesting that these trip purposes are more frequently associated with inefficient trips.
2. **On weekends, inefficient trips peak at 13:00 (23.1%) and 15:00 (14.8%). Errand/Supplies and Meeting** are the most common trip purposes during these periods,suggesting that business-related travel remains significant even on weekends and contributes to a higher proportion of inefficient trips.